

AI Enterprise Operations Intelligence Platform

Complete JavaFX Frontend Specification Document

WFM Pillar — Enterprise Frontend Architecture & Wireframe Specification

DOCUMENT PURPOSE

This document defines the complete frontend architecture, UI structure, wireframe implementation approach, realtime synchronization architecture, communication architecture, CSV ingestion system, visualization requirements, and operational workflow structure for the JavaFX-based WFM platform.

This frontend specification is strictly aligned with:

- WFM_Pillar_Wireframe_Specification_v2.0
- WFM × Finance API Route Documentation
- WFM SAAS Functionalities Reference

The purpose of this document is to ensure that the JavaFX frontend is implemented as a direct enterprise-grade rendering of the provided WFM wireframes without changing the original structure.

CORE IMPLEMENTATION RULE

The WFM_Pillar_Wireframe_Specification_v2.0 document is the primary UI blueprint.

The frontend implementation must preserve:

- exact layout hierarchy
- exact panel ordering
- exact dashboard structure
- exact workflow arrangement
- exact operational density
- exact table positioning
- exact KPI arrangement
- exact forecasting layouts
- exact scheduling layouts

- exact RTM structures
- exact enterprise workflow behavior

The frontend must NOT:

- redesign the UI
- modernize the UI structure
- simplify layouts
- rearrange panels
- convert the design into SaaS-style dashboards
- create oversized card-based layouts
- introduce consumer-style spacing

The JavaFX frontend acts only as:

- rendering layer
- realtime layer
- interaction layer
- synchronization layer
- visualization layer

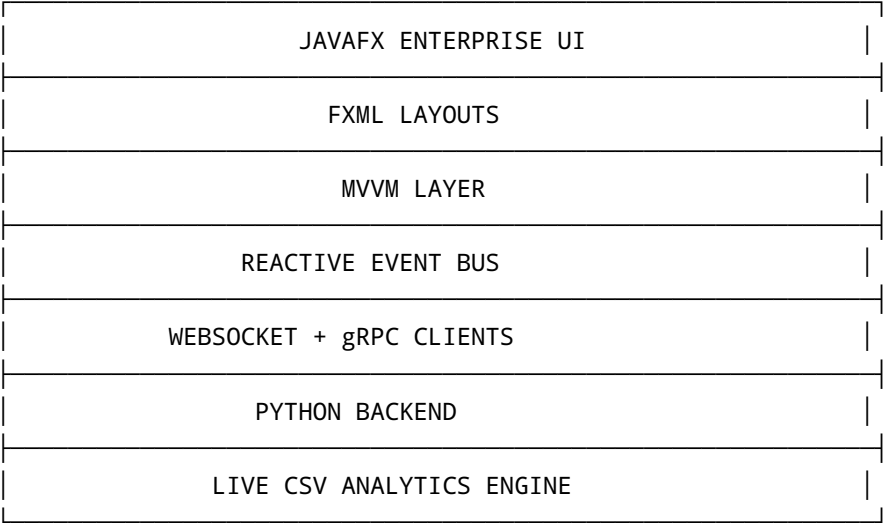
while preserving the original wireframe structures.

FRONTEND TECHNOLOGY STACK

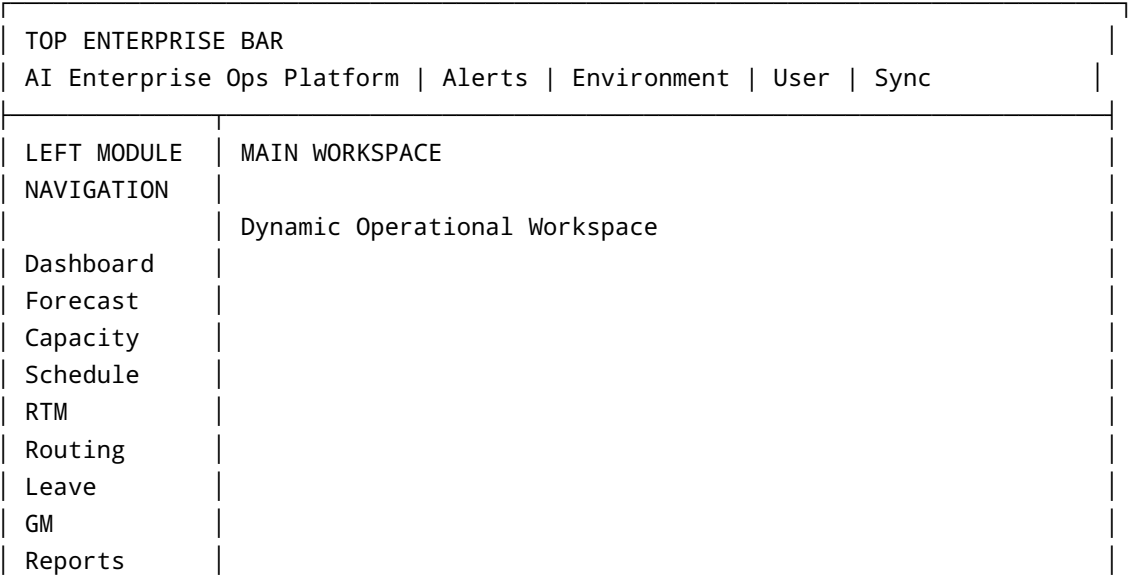
Layer	Technology
Frontend Framework	JavaFX
Architecture Pattern	MVVM
UI Definition	FXML
Styling	JavaFX CSS
Backend Communication	gRPC
Realtime Updates	WebSocket
Charts	ChartsFX
Animation Engine	JavaFX Transitions + AnimateFX
Notifications	ControlsFX
Icons	Ikonli
Reactive Streams	ReactFX
Table Virtualization	VirtualFlow

Layer	Technology
Theme System	Enterprise Dark Theme

HIGH-LEVEL FRONTEND ARCHITECTURE



GLOBAL APPLICATION LAYOUT



Finance	
Operations	
Audit	
GLOBAL STATUS BAR	
WebSocket ✓	gRPC ✓ CSV Watcher ✓ Sync ✓ Queue ✓

The computational chain must remain globally visible across all modules.

The spine must:

REALTIME CSV DIRECTORY ARCHITECTURE

Dedicated CSV Directory

A dedicated backend directory will store all CSV files.

Example:

```
/data/live_csv/
```

The directory must be monitored continuously.

REQUIRED LIVE BEHAVIOR

Whenever:

- a new CSV file is added
- a CSV file is modified
- a CSV file is deleted
- a CSV file is renamed

The platform must automatically:

1. detect the file change
2. validate the CSV
3. process analytics
4. recalculate metrics
5. update forecasting
6. update operations analytics
7. update finance calculations
8. broadcast live events
9. refresh JavaFX UI automatically

without restarting the application.

CSV MONITORING FLOW

```
CSV DIRECTORY
```

```
↓
```

```
Python Watchdog Observer
```

↓
Validation Layer
↓
Analytics Processing Engine
↓
Realtime Event Dispatcher
↓
WebSocket Broadcast
↓
JavaFX UI Refresh

FRONTEND LIVE UPDATE FLOW

WebSocket Event
↓
Reactive Event Bus
↓
Observable Models
↓
Charts / Tables / KPIs Updated
↓
Smooth Animated UI Refresh

COMMUNICATION ARCHITECTURE

gRPC RESPONSIBILITIES

gRPC will be used for:

- forecasting calculations
- capacity planning calculations
- finance calculations
- scheduling optimization
- report retrieval
- historical analytics
- AI recommendation retrieval
- operational calculations
- GM estimation
- audit retrieval

WEBSOCKET RESPONSIBILITIES

WebSockets will be used for:

- live KPI updates
- CSV change events
- RTM updates
- occupancy updates
- SLA updates
- adherence updates
- operational alerts
- queue monitoring
- dashboard updates
- chart refreshes
- realtime notifications

CSV UPLOAD MODULE

The frontend must support CSV upload.

CSV UPLOAD FLOW

```
graph TD; A[JavaFX Upload] --> B[gRPC Upload API]; B --> C[Python Validation Layer]; C --> D[Store Into Live CSV Directory]; D --> E[Watcher Detects File]; E --> F[Analytics Recalculation]; F --> G[WebSocket Refresh]; G --> H[Live UI Update];
```

CSV UPLOAD UI

CSV Upload
[Drag & Drop CSV Here]
OR
[Browse CSV]
Validation Status
Processing Status
Upload Progress
Synchronization Status

MODULE 1 — WFM DASHBOARD

The dashboard structure must exactly replicate the wireframe specification.

WFM DASHBOARD	[23 May 2026]	[LOB: All]	[Site: All]	[Refresh]
KEY HEALTH INDICATORS (click any card to drill into module)				
REQ COVER 98.4% Target 100% ▲ Trending	SCH EFFIC 91.2% Target ≥75% ✓ On track	ADHERENCE 95.8% Target ≥95% ✓ On track	ATTRITION 8.1% ann Prev: 7.8% △ Rising	
FCA 87.3% Prev:84.1% ✓ Improving	OCCUPANCY 78.4% Target <85% ✓ Healthy	IN-OFF SH 12.6% Budget 12% ✓ On track	GM EST 34.8% Budget 35% △ -0.2pp	

6-MONTH COVERAGE TREND	AI RECOMMENDATIONS (Explainable)
[Line chart]	△ W26 coverage projected at 93.2%
Req Production Hrs	→ Recommend opening 4 hires now
Cap Production Hrs	[Accept] [Modify] [Dismiss]
Actuals	△ Attrition in LOB-2 rising
W23 W24 W25 W26 W27 W28	→ 3 CEs flagged pre-leave pattern
98% 102% 95% 93% -- --	[View Profiles] [Alert Ops]
CROSS-DEPARTMENT LIVE FEED	
OPS SL today: 81.2% ✓ AHT: 318s vs 320s est	
FINANCE Direct Cost/PPH: €10.39 GM gap: -0.2pp △	
RECRUITMENT 4 pending hire approvals W26 NHT confirmed ✓	
[Go to OPS] [Go to Finance] [Go to Recruitment]	

DASHBOARD FEATURES

Required Dashboard Features

- realtime KPI updates
- animated KPI counters
- live charts
- websocket synchronization
- realtime AI recommendations
- operational alerts
- drilldowns
- live cross-department feed
- synchronized analytics updates

MODULE 2 — FORECAST STUDIO

MODULE PURPOSE

Forecast Studio is responsible for:

- forecast creation
- version management

- forecast locking
- forecasting comparisons
- AHT learning curve management
- event impact forecasting
- forecasting governance
- forecast-to-capacity integration

The module must preserve the exact layout and operational structure defined in the WFM specification.

FORECAST STUDIO — COMPLETE STRUCTURAL WIREFRAME

FORECAST STUDIO

Type:[Locked ▼] Period:[Jun-Nov 2026 ▼] LOB:[All ▼]

Channel:[Phone ▼] Granularity:[Weekly ▼] View:[Table][Chart]

ASSUMPTIONS TABLE – LOCKED FORECAST

Week	Offered	Handled	AHT	Avail	In-Off	Out-Off	Leavers
W23	12,500	12,125	320	31.2%	12.0%	4%/18%	2.0
W24	13,200	12,804	317	30.8%	11.8%	4%/17%	1.5
W25	14,100	13,677	315	31.5%	12.2%	4%/19%	3.0 ▲
W26	13,800	13,386	318	31.0%	12.0%	4%/18%	1.0

▲ W25 Leavers above trend. Verify with Operations.

[Edit Row] [Add Event] [Lock Week] [Export CSV] [Send to Cap Plan]

FORECAST VERSION COMPARISON

Version	W23	W24	W25	W26
Client Locked Forecast	12,500	13,200	14,100	13,800
Previous Internal Forecast	12,200	12,950	13,800	13,500
Current Internal Forecast	12,125	12,804	13,677	13,386
Actuals	12,048	12,711	—	—
FCA (Client)	90.4%	87.3%	—	—
FCA (Internal)	96.2%	96.2%	—	—

AI: Client forecast running +3.8% above actuals consistently.

Recommend internally agreed forecast at -4% vs client locked for W26+.											
[Accept AI Recommendation] [Set Custom Adjustment] [Keep Client Volume]											
AHT LEARNING CURVE MANAGER											
Prod Week	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W12+
AHT (s)	1200	1020	867	737	626	595	559	509	475	459	450
Group FTEs Prod Week AHT Weighted Hrs Contribution											
EVENT IMPACT REGISTER											
Event Type		Start		End		Impact %		Applied To			
Summer Holiday		04-Aug		04-Aug		-35%		W32			
Product Launch		12-Jun		18-Jun		+22%		W24			
Black Friday		27-Nov		01-Dec		+45%		W48			
FORECAST ROLLING PERIOD STATUS											
Finance Milestone			Due Date			Status					
Q3 Reforecast			30-Jun-2026			Open					
Q4 Reforecast			Sep-2026			Upcoming					

FORECAST STUDIO — LIVE DATA REQUIREMENTS

The Forecast Studio module must support realtime synchronization.

Whenever:

- forecast assumptions change
- CSV data changes
- operational actuals change
- AHT changes
- event registers update
- forecast versions are modified

the module must refresh automatically.

FORECAST STUDIO — WEBSOCKET EVENTS

Example websocket events:

```
{
  "event": "FORECAST_UPDATED",
  "week": "W26",
  "field": "offered_volume",
  "old_value": 13800,
  "new_value": 14200
}
```

```
{
  "event": "AHT_CURVE_UPDATED",
  "group": "New Hires",
  "week": "W2"
}
```

FORECAST STUDIO — gRPC RESPONSIBILITIES

The Forecast Studio module will use gRPC for:

- forecast calculations
- forecast persistence
- version comparison
- FCA calculations
- learning curve calculations
- event impact recalculation
- operational forecasting analytics

FORECAST STUDIO — CHART REQUIREMENTS

Charts must support:

- smooth transitions
 - realtime updates
 - animated overlays
 - zoom
 - pan
 - synchronized hover states
 - actual vs forecast comparison
 - interval highlighting
-

FORECAST STUDIO — TABLE REQUIREMENTS

Tables must support:

- inline editing
- virtualization
- sticky headers
- live updates
- filtering
- sorting
- bulk paste
- AI warnings
- operational highlighting

FORECAST STUDIO — JAVA FX STRUCTURE

```
ForecastStudio.fxml
|
├─ ForecastToolbar
├─ AssumptionsTable
├─ VersionComparisonPanel
├─ AHTLearningCurvePanel
├─ EventImpactRegister
├─ ForecastRollingStatus
└─ ActionButtonsRow
```

FORECAST STUDIO — UI RULES

The Forecast Studio must preserve:

- dense operational layout
- compact spacing
- enterprise workflow arrangement
- horizontal analytical sections
- table-first operational structure
- compact forecasting analytics design

without redesigning the structure.

MODULE 2 — FORECAST STUDIO

FORECAST STUDIO
Type:[Locked ▼] Period:[Jun-Nov 2026 ▼] LOB:[All ▼]
Channel:[Phone ▼] Granularity:[Weekly ▼]
ASSUMPTIONS TABLE – LOCKED FORECAST
FORECAST VERSION COMPARISON
AHT LEARNING CURVE MANAGER
EVENT IMPACT REGISTER
FORECAST ROLLING PERIOD STATUS

FORECAST STUDIO FEATURES

Forecast Features

- editable forecast tables
- version comparison
- live forecast recalculation
- forecast overlays
- AHT learning curve visualization
- AI forecast recommendations
- realtime analytics synchronization
- animated forecast graphs
- operational alerts

MODULE 3 — CAPACITY PLANNER

REQUIREMENT BUILD ENGINE
CAPACITY BUILD ENGINE

SIX-MONTH ROLLING COVERAGE DASHBOARD
LEAVERS TRACKER
STAKEHOLDER SIGN-OFF WORKFLOW
RECRUITMENT TRACKER
COMPANY RESOURCE FORECAST

CAPACITY PLANNER FEATURES

- realtime coverage analysis
- hiring alerts
- capacity forecasting
- leaver projections
- stakeholder workflow management
- interactive requirement calculations
- coverage trend visualization
- AI staffing recommendations

MODULE 4 — SCHEDULE BUILDER

SCHEDULE BUILDER
INTERVAL COVERAGE TABLE
INTERVAL-LEVEL COMPLIANCE
WEEKLY SCHEDULING METRICS SUMMARY
SCHEDULING METHODS SUPPORTED
NEW HIRE SCHEDULING SUB-MODULE

REQUIRED INTERVAL COVERAGE TABLE

Interval
REQ
SCH
ACT
GAP
Coverage Rate
Status
AI Action Suggested

SCHEDULE BUILDER FEATURES

- drag-and-drop scheduling
- OT suggestions
- VTO suggestions
- interval heatmaps
- live coverage analysis
- schedule optimization
- realtime staffing updates
- compliance tracking

MODULE 5 — RTM CONSOLE

RTM CONSOLE	
LIVE KPI SNAPSHOT	
ACTION MATRIX	
ADHERENCE GRID	
ACTION LOG	
EOD SERVICE LEVEL PROJECTION	

RTM FEATURES

- realtime SLA updates
 - occupancy monitoring
 - queue monitoring
 - adherence tracking
 - action matrix recommendations
 - operational alerts
 - interval projections
 - websocket-driven updates
-

MODULE 6 — ROUTING MANAGER

ROUTING MANAGER
QUEUE MANAGEMENT
SKILL GROUPS
ROUTING RULES
OVERFLOW MANAGEMENT
LIVE ROUTING HEALTH

MODULE 7 — LEAVE PLANNER

LEAVE PLANNER
LEAVE CALENDAR
LEAVE REQUEST QUEUE
APPROVAL PANEL
COVERAGE IMPACT ANALYSIS

AI LEAVE RECOMMENDATIONS

MODULE 8 — GM ESTIMATOR

GM ESTIMATOR

GM STAGES

WHAT-IF ANALYSIS

COST PROJECTIONS

ANNUALISED IMPACT

FINANCE ALERTS

MODULE 9 — REPORTS CENTRE

REPORTS CENTRE

REPORT FILTERS

REPORT GRID

EXPORT CONTROLS

SCHEDULED REPORTS

FINANCE MODULE

DAILY COST BREAKDOWN
MONTHLY ROLLUP
OT PREVIEW
PAY COMPONENT VISUALIZATION
POLICY ALERTS

OPERATIONS MODULE

PRODUCTIVE HOURS
SHRINKAGE ANALYSIS
OCCUPANCY ANALYSIS
COVERAGE MONITORING
TREND ANALYSIS
ANOMALY DETECTION

VISUALIZATION REQUIREMENTS

REQUIRED GRAPH TYPES

Visualization	Purpose
Line Charts	Forecasting

Visualization	Purpose
Area Charts	Occupancy
Heatmaps	Interval Coverage
Gauges	SLA
Timelines	Scheduling
Stacked Bars	Shrinkage
Sankey Diagrams	Capacity Flow
Scatter Plots	Anomaly Detection

GRAPH REQUIREMENTS

All visualizations must support:

- smooth transitions
- realtime updates
- hover tooltips
- zoom
- pan
- animated refresh
- synchronized filtering
- interval highlighting
- operational overlays

FRONTEND PACKAGE STRUCTURE

```
frontend-javafx/
|
├─ app/
├─ grpc/
├─ websocket/
├─ modules/
├─ components/
├─ services/
├─ models/
├─ utils/
├─ themes/
```

```
|— assets/  
|— fxm1/
```

THEME REQUIREMENTS

Enterprise Theme Design

Element	Design
Background	Deep charcoal
Panels	Dense enterprise borders
Accent	Neon blue
Success	Emerald
Warning	Amber
Critical	Red
Typography	Compact enterprise typography
Layout	Operational density
Charts	Smooth gradients

PERFORMANCE REQUIREMENTS

The frontend must support:

- high-frequency realtime updates
- smooth animations
- large tables
- large CSV ingestion
- low-latency synchronization
- websocket reconnect handling
- scalable operational analytics
- responsive enterprise workflows

GOVERNANCE REQUIREMENTS

The frontend must support:

- audit visibility
 - operational approvals
 - change history
 - override tracking
 - role-based access
 - workflow authorization
 - operational compliance monitoring
-

FINAL IMPLEMENTATION PRINCIPLE

The frontend must be implemented as:

Exact JavaFX Rendering
of the WFM Wireframe Specification
with Realtime Enterprise Analytics

The specification document defines:

- structure
- layout
- hierarchy
- workflow
- panel arrangement
- operational composition

while JavaFX adds:

- realtime behavior
- smooth animations
- websocket streaming
- gRPC integration
- interactive visualizations
- CSV synchronization
- enterprise responsiveness
- live operational analytics